

SmartMatch: An Adaptive NLP-Driven Platform for Intelligent Connections

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Abstract—The SmartMatch platform stands as a flexible NLP-based system which enables automatic matching between various domains of individual profiles and opportunities. Users need to produce complete profiles that contain resumes together with keywords and descriptions of their skills and research interests. Job descriptions and required certifications and relevant keywords come from Organizations, recruiters or collaborators. The sophisticated NLP methods present in the platform analyze and compare profiles to matching opportunities by evaluating semantic points so results become both exact and quick. SmartMatch demonstrates capabilities which go beyond job portal applications because it extends toward academic linking opportunities for project assignment and professional network creation across scalable use cases. The platform improves both decision-quality and workflow automation through its ability to be flexible and precise.

Index Terms—SmartMatch. Profile matching, NLP techniques, academic collaborations, conference management.

I. INTRODUCTION

Nowadays, while the academic world moves at great speeds, the process of conducting conferences and delegating them to appropriate reviewers for the received manuscript is a prerequisite that requires a lot of consideration. The manual execution of the reviewer assignments is usually associated with some inefficiencies and possible biases as well as the delay in peer review proceedings. The SmartMatch system overcomes these issues with a system that automates and infuses intelligence to simplify paper submission, reviewer management, and even assignment operations, which increases transparency and accuracy in review workflow.

SmartMatch is a full-stack web application that comprises the MERN stack— React.js for frontend while Node.js/Express.js form the backend and the database is the PostgreSQL. The system provides for several user roles, including the authors, reviewers, and administrators, with specialized dashboards. Authors can upload papers, check their statuses and reviewers can view assigned submissions and review them, while administrators can track the whole ecosystem, manage users and supervise matching between the reviewers and the papers.

Its intelligent algorithm for assignment of reviewers to papers is one of the main features of SmartMatch, which vectorizes both abstracts of papers and profiles of reviewers via NLP-based techniques. The similarity between these vectors is measured using cosine similarity meaning that every paper is matched to reviewers who are very knowledgeable about the subject. Moreover, this strategy does not only enhance the relevance of reviews but also reduces conflicts of interests and overburdening of particular reviewers.

Outside basic functionality, SmartMatch includes basic security features such as email verification, OTP password resets, and role-based authentications. The system is built for scalability and capability of being scaled up to conferences of all sizes and scopes. With SmartMatch, reviewer assignment is automated and conference workflows are simplified, thus offering a strong solution to organizers who would like to manage the peer review process with more efficiency, fairness and preciseness.

II. LITERATURE SURVEY

The research study from Zhao et al. [1] examines peer reviewer scheduling systems from the previous thirty years while highlighting the automation requirements for the peer review system. The author explains how NLP techniques enhance reviewer-paper matching precision because this matching represents the core of fair peer review procedures. Zhao delivers an extensive overview of research algorithms for RAP together with evaluation methods and potential research paths for the field. Collaborative reviewer-paper matching receives attention through Anjum et al. [2] proposed approach that resolves previous systems' vocabulary mismatches and incomplete topic matches. The use of word embedding allows the model to detect effective submission-reviewer relevance fast and efficiently. The experimental findings indicate that the system delivers enhanced matching precision that exceeds performances of current advanced approaches. Aksoy et al. [3] reviewed the Reviewer Assignment Problem systematically by examining different methods for enhancing assignment accuracy through optimization techniques and AI-driven models as well as heuristic approaches. The paper investigates important

obstacles which include fairness aspects alongside challenges related to obtaining expertise agreements and dealing with complex computational needs. The latest developments have produced more efficient reviewer-paper matching methods that resolve issues with bias and scalability limitations during peer-review operations.

Pendyala et al. [4] designed a machine learning platform that assists in the peer review phase for research paper acceptance procedures. The system employs SVM and LSTM and BERT and PaLM 2 models to forecast acceptance decisions and assign review scores and select reviewers. The experimental findings confirm DistilBERT’s success for acceptance forecasting at 86% accuracy alongside enhancements to peer review operations that did not substitute judgment capabilities and reveal potential innovations for enhancing explainability together with model deployment methods. The hybrid features matching system proposed by Khilji et al. [5] uses BERT-based NLP with Jaccard similarity to calculate efficient feature similarities. The method minimizes the time needed for human-based features correlation analysis or alternative techniques. This method provides effective combined performance while the study indicates better results can be achieved with extra similarity metrics but at the expense of efficiency. Ali et al. [6] created an automated Resume Classification System (RCS) through NLP and ML techniques to handle job application sorting tasks more efficiently. A study evaluates ML algorithms where the SVM classifier achieved the best performance through its classification of resumes with more than 96% accuracy. TF-IDF and other NLP extraction methods used for feature extraction prove highly effective in enhancing automated resume classifier accuracy according to research findings.

Watanabe et al. [7] present a model to handle academic conference paper-to-reviewer assignment that integrates both preference-based and topic-based assignment strategies. The method uses expert review knowledge from past authorships to optimize assignment processes. Review feedback can serve as a basis for dynamic reviewer reassignment which promotes better assignment quality according to research. Boddu et al. [8] suggest an automated resume screening method which combines NLP-based Resume Summarizer software with Job Description Matcher software. Recruitment efficiency improves through their system that applies OCR combined with text preprocessing and machine learning capabilities. The research findings establish that automated recruitment techniques through NLP create new opportunities for industry-wide hiring process optimization. Narendra and Hashwanth [9] developed an NLP-based solution for automatic resume screening through entity extraction followed by entity ranking. The use of spaCy technology enables this method to parse documents effectively and categorize them which means less time needs to be spent doing manual resume evaluations. This system assists recruitment professionals to perform fast candidate selection.

TABLE I
SUMMARY OF RELATED WORKS

Paper	Objective	Techniques Used	Key Findings	Limitations
[1] Zhao and Zhang, 2022	Automate reviewer assignment processes	Optimization, machine learning, fairness models	Address fairness, scalability, and bias	Limited focus on AI fairness integration
[2] Anjum et al., 2023	Address mismatches in reviewer matching	Word embeddings, NLP topic modeling	Improved matching precision, reduced bias	Needs large-scale dataset testing
[3] Aksoy et al., 2023	Systematic review of RAP approaches	Optimization, AI, heuristic techniques	Enhanced fairness and scalability methods	Limited real-time RAP application exploration
[4] Pendyala et al., 2023	Automate research review using NLP	LLMs, transformers, NLP pipelines	Improved categorization with cutting-edge NLP	High computational resource dependency
[5] Khilji, 2023	Enhance NLP-based feature matching	TF-IDF, multi-domain classification	Strong domain-specific matching precision	Lacks advanced matching methodologies
[6] Ali et al., 2022	Automate resume classification with NLP	TF-IDF, SVM, Naïve Bayes, KNN	Achieved 96% classification accuracy	Needs broader industry evaluation

III. METHODOLOGY

A. Dataset Details

The dataset that has been used to evaluate the SmartMatch system consists of 50 research papers gathered from five different domains altogether. Machine Learning (ML), Deep Learning (DL), Cryptography, Natural Language Processing (NLP), and Computer networks, with 10 papers taken for each of these domains. Each paper has essential metadata like title, abstract, keyword and domain classification, of which, the abstracts are the main source of material for vectorization and similarity measures. Correspondingly, a pool of five reviewers was developed, each specializing in a respective one of the five domains. These reviewer profiles were built based on descriptions of domain-specific expertise and vectorized to make it possible for semantic comparison with submitted papers. Such a structured dataset allowed SmartMatch system to simulate real-world conference scenarios and test its reviewer-paper matching algorithm by means of cosine similarity, thus achieving reliable, relevant-to-the-domain, and balanced reviewer assignment.

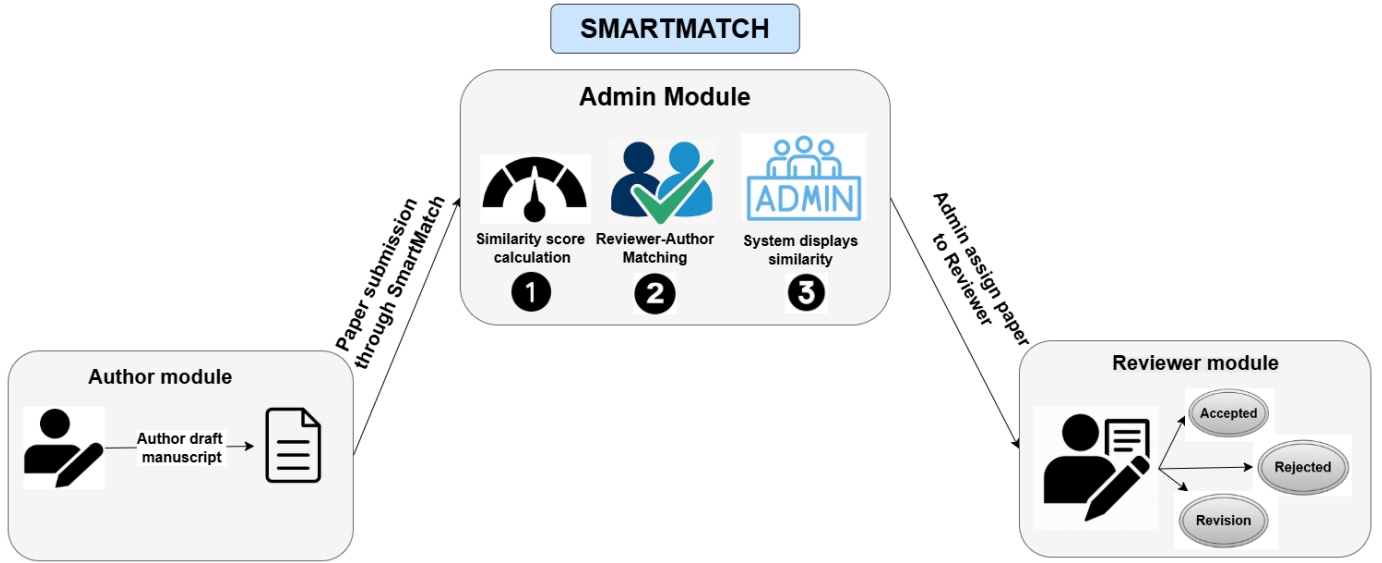


Fig. 1. Architecture Diagram

B. System Architecture

The system architecture of SmartMatch is based on a modular and role-based structure that includes three main components. Author Module, Admin Module and Reviewer Module. The workflow starts when the author uploads a manuscript from the Author Module. After submission, the Admin Module is the one that has a task to process the paper as it calculates the similarity score between the manuscript's content and reviewer expertise vectors based on natural language processing techniques. Based on these scores the system will be able to intelligently match the reviewers-authors, and present the similarity metrics to the administrator to verify the results. The administrator then gives the paper to the most suitable reviewer. The Reviewer Module allows the assigned reviewer to gain access to the manuscript and give a decision- Accepted, Rejected, or Revision Required. This modular architecture, as illustrated in Fig. 1, facilitates an automated and efficient review workflow that improves the process of assigning reviewers as well as the accuracy of the reviewer assignment while maintaining transparency and fairness in peer review management.

C. Embedding Vector Generation

Generation of embedding vector-based embedding vector generation was an important aspect of the SmartMatch system towards allowing semantic comparison between submitted papers and reviewer expertise. This procedure involved the process of transforming the textual data like author abstracts and reviewer keywords or expertise descriptions into numerical vectors that can be used in similarity calculation. A few methods that were used for this purpose were two popular ones: TF-IDF (Term Frequency-Inverse Document Frequency), and GloVe (Global Vectors for Word Representation).

TF-IDF is a measure in statistics which ranks the relevance of a word in a document of a set. It's working way counts

the frequency of a word in a document (term frequency) and divides it by the frequency of the same word throughout all documents (inverse document frequency). This leads to sparse vector representation as more meaningful and distinctive words in the context of the given document are given more weight. TF-IDF was used in SmartMatch to represent both the relevance of keywords and abstract terms to the authors and the reviewers, thus enabling a lightweight but efficient computation of similarity scores.

On the other hand GloVe is a pre-trained word embedding model which embeds the semantic meaning of the words based on the global word-word co-occurrence statistics of the large text corpus. In contrast to TF-IDF, the GloVe method will produce dense vector representations with a fixed length in which semantically similar words are mapped into similar points of the vector space. In SmartMatch, GloVe embeddings were then averaged across the abstract or reviewer text to give rise to document-level vectors that capture a deeper understanding of the context beyond raw word frequency.

Applying TF-IDF and GloVe to the system enabled a strong base for creating relevant embedding vectors that were then used in calculating cosine similarity scores between papers and reviewer profiles. This back-and-forth embedding approach improved the precision and topicality of the matching between reviewers and papers in disparate areas.

D. Reviewer Matching Algorithm

Algorithm 1 assigns a submitted paper with the most appropriate reviewer as per cosine similarity on vectorized representations. First, abstract of the paper as well as all reviewer profiles are converted into a vector of embeddings by means of a language model. Such high-dimensional vectors are then projected to a low dimension space using UMAP for the ease of computation. To determine alignment, cosine similarity between paper vector and each reviewer vector

is computed. Reviewers are placed in order of decreasing similarity. The highest-matching reviewer is chosen, making sure that they have not reached the predefined limit of the paper assignment (for example, 3 papers). If so, the algorithm then goes to the next highest scorer until a match occurs.

Algorithm 1 Reviewer Matching Based on Cosine Similarity

Require: Paper abstract P , Reviewer Profiles $R = \{R_1, R_2, \dots, R_n\}$

Ensure: Matched Reviewer R^* for paper P

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1:  $v_P \leftarrow \text{Embed}(P) \triangleright$  Convert paper abstract to embedding vector
2: for  $i = 1$  to  $n$  do
3:    $v_{R_i} \leftarrow \text{Embed}(R_i)$ 
4: end for
5:  $v_P^{\text{UMAP}} \leftarrow \text{UMAP}(v_P)$ 
6: for  $i = 1$  to  $n$  do
7:    $v_{R_i}^{\text{UMAP}} \leftarrow \text{UMAP}(v_{R_i})$ 
8: end for
9: for  $i = 1$  to  $n$  do
10:   $\text{sim}_i \leftarrow \frac{v_P^{\text{UMAP}} \cdot v_{R_i}^{\text{UMAP}}}{\|v_P^{\text{UMAP}}\| \cdot \|v_{R_i}^{\text{UMAP}}\|} \triangleright$  Cosine similarity
11: end for
12:  $R_{\text{sorted}} \leftarrow \text{sort}(R, \text{sim}) \triangleright$  Sort reviewers by similarity (descending)
13: for each  $R_j$  in  $R_{\text{sorted}}$  do
14:   if  $\text{PapersAssigned}(R_j) < 3$  then
15:      $R^* \leftarrow R_j$ 
16:     return  $R^*$ 
17:   end if
18: end for
19: return No Reviewer Available

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IV. RESULTS AND DISCUSSION

A. Login And Signup Page

The SmartMatch platform contains a secure and convenient interface for authenticating users – a set of dedicated pages for Sign Up and Login like in Fig. 2 and Fig. 3 respectively. These pages are built to deliver a seamless onboarding, strong access control based on the role of a user that will be either an Author, Reviewer, or Admin.

The Sign Up page has a space where new users can register by filling in details such as username, email, password and first, last names, date of birth, organization and country. In order to improve security and avoid automatic registrations, captcha verification feature is incorporated. Also, real-time validations are provided for the confirmation of password, and for the mandatory fields to maintain the form integrity before the submission of the same.

Through the Login page (Figure 1), returning users are able to access the platform, using their username and password. It contains toggling functionality for password visibility as well as links to recover lost password and register a new user. After logging in successfully, the users are redirected to role-specific dashboards, which incorporates personalized and allowing

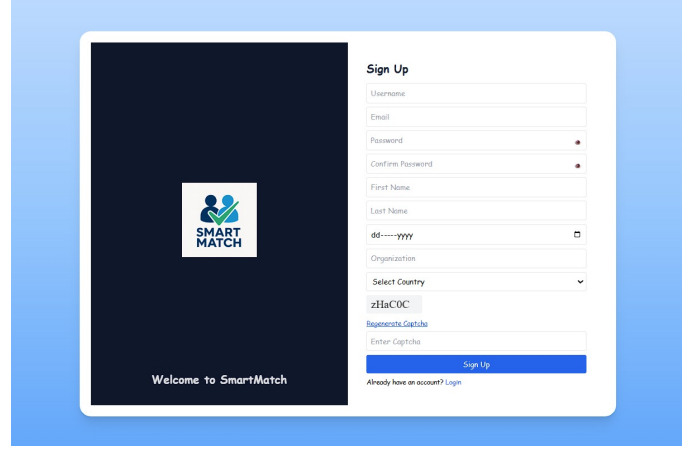


Fig. 2. Signup Page

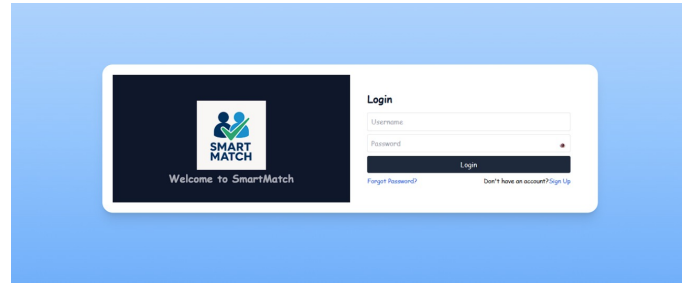


Fig. 3. Login Page

setting access restriction environment. The interface gets styled up with React.js and Tailwind CSS to ensure responsiveness on different devices and consistent brand identity with the SmartMatch brand.

B. User Dashboard

SmartMatch system offers an interactive user-oriented User Dashboard as shown in the Fig. 4, intended to facilitate the paper submission workflow for authors. When a user successfully logs in he/she is welcomed by a summary panel that shows vital statistics such as Total Submissions, Accepted Papers, Accepted with Revision and Rejected Papers. This allows the users to view the status of their submissions from a single screen. Underneath the statistics panel, there is a table

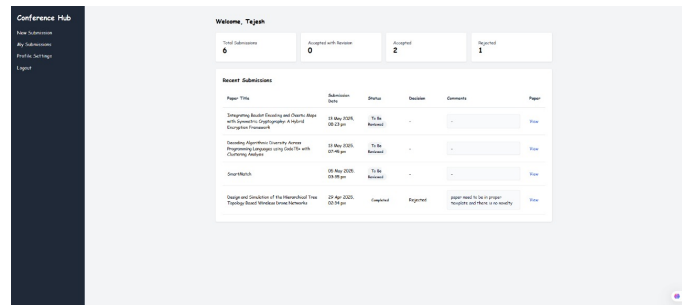


Fig. 4. User Dashboard

of Recent Submissions on the dashboard which exhibits the details of each submitted manuscript. The columns available in the table include Paper Title, Submission Date, Status, Decision, Reviewer Comments, and View link for viewing the uploaded PDF. The status field reports on the prevailing stage of review (e.g., “To Be Reviewed” or “Completed”), and the newly added decision and comments fields are updated from reviewer feedback to provide transparency and up-to-date information to the author.

Left sidebar provides the navigation options like New Submission, My Submissions, Profile Settings, and Logout that makes the interaction with various platform’s functionalities effortless. The entire dashboard is developed using React.js and Tailwind CSS, which makes it responsive and modern and provides usability benefits and improved accessibility. This dashboard enables the authors to effectively monitor the submission and they are kept updated on the reviewing process.

C. Reviewer Dashboard

The SmartMatch system has an integrated Reviewer Dashboard as shown in Fig. 5, which aims to help reviewers to efficiently deal with their assigned submissions. Upon logging in the reviewers are greeted with a summary of some of the main statistics such as Total Assigned, Under Review to name only a few. This high level overview allows reviewers to get a quick gist of their amount of work and prioritize them as well.

At the center of the dashboard, we have the recently assigned papers which give vital details such as Paper ID, Paper Title, Authors, Due Date, PDF Access, Review Status, Decision, and Reviewer Comments. Reviewers are able to directly access the manuscript, choose correct decision (e.g. Accept, Reject, Revision) and leave detailed comments from the single interface. Save button is made available for each entry to edit and submit the review in real time.

Under the assignment section, an Upcoming Deadlines panel appears that openly displays papers approaching their review periods, ensuring that such papers receive timeliness in responses. In addition, a Review Progress tracker keeps tally of completed reviews and rate of on time submission which works to maintain the commitment and transparency in the review process.

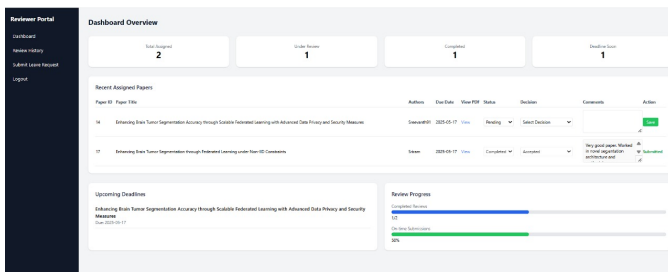


Fig. 5. Reviewer Dashboard

D. Admin Dashboard

Fig. 6 shows the Admin Dashboard of the SmartMatch system is an overall control panel for adding, editing, and deleting the components of the system (users, reviewers, paper submissions, and reviewer-paper assignments). When administrators gain access to the dashboard, they receive the information regarding all the important metrics of the system, such as the number of Active Users, Total Reviewers, Pending Assignments, and the Login Activities records. These metrics give a holistic overview of system engagement and how to track use of platform.

The Paper Assignment Panel is one of the key functions of the Admin Dashboard, and consists of a panel of submitted papers and a dynamically generated list of Suggested Reviewers. For each one of the reviewers suggested, there is a match score indicating the cosine similarity between the abstract of paper and the expertise profile of the reviewer. Such an intelligent matching mechanism helps the administrators to make informed decisions when assigning reviewers. The “Assign” action allows the admin to complete the assignment manually depending on relevance and availability. The

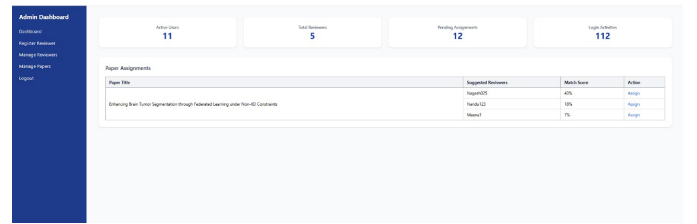


Fig. 6. Admin Dashboard

other than assignment operations, the admin is equipped with options to register new reviewers, the management of the reviewer accounts, and tracking the paper submissions through sidebar menu. The dashboard lay-out is coded with React.js and Tailwind CSS to ensure a smooth and responsive interface.

E. Comparative Analysis of different Embeddings

The Table II includes a comparative analysis of TF-IDF and GloVe embeddings on the basis of true positives and false positives with respect to five research domains. Machine Learning (ML), Deep Learning (DL), Cryptography, Networks, Natural Language Processing (NLP). From the above observation, TF-IDF always produces more true positive and less false positives in all domains and hence better precision and accurate reviewer-paper matches. For example, in Networks and NLP, TF-IDF obtained 9 true positives but only 1 false positive. On the other hand, GloVe embeddings had a lower number of true positives and a higher number of false positives, which was especially true for NLP domain, it only identified 4 true positives and 6 false positives. Based on this, we can conclude that the TF-IDF embeddings work better to represent domain-specific relevance in SmartMatch system reviewer assignment.

TABLE II
COMPARISON OF TRUE POSITIVES AND FALSE POSITIVES FOR TF-IDF
AND GLOVE EMBEDDINGS

Embedding	Domain	True Positive	False Positive
TF-IDF	ML	8	2
	DL	8	2
	Crypto	7	3
	Networks	9	1
	NLP	9	1
GloVe	ML	6	4
	DL	5	5
	Crypto	6	4
	Networks	6	4
	NLP	4	6

V. CONCLUSION AND FUTURE WORK

The SmartMatch system is a holistic and intelligent solution to automate the assignment process of reviewers in an academic conference. By leveraging the latest web technologies along with the methods of natural language processing, the platform manages to facilitate the entire peer review workflow without sacrificing anything. By applying TF-IDF and GloVe embeddings for vector representation of paper abstracts and expertise of reviewers, it is possible to perform semantic matching with the help of cosine similarity. From the experimental results of the paper, it is shown that TF-IDF has a greater capacity to pick domain-specific relevance and achieve higher accuracy in multiple research areas. The system offers separate dashboards for the authors, reviewers, and administrators wherein papers can be submitted smoothly, tracked, and decision can be managed in a secure and convenient environment. Regarding the future, SmartMatch could be improved with the addition of powerful language models like BERT or Sentence-BERT for better quality of embeddings and understanding the context. Future versions can be also equipped with the workload balancing for reviewers, the detection of conflict of interests, automatic plagiarism check to ensure fair play and integrity. Other features such as real-time deadline alerts, reviewer performance metrics, and multi-domain expertise handling can make a huge difference in usability and scalability. Lastly, implementing the system on an online (cloud) infrastructure with portable support would allow mass spread for real-world academic conferences, and SmartMatch would become an enormously extensible, future-ready peer review management instrument

REFERENCES

- [1] X. Zhao and Y. Zhang, "Reviewer assignment algorithms for peer review automation: A survey," *Information Processing & Management*, vol. 59, no. 5, p. 103028, 2022.
- [2] O. Anjum, H. Gong, S. Bhat, W.-M. Hwu, and J. Xiong, "Pare: A paper-reviewer matching approach using a common topic space," *arXiv preprint arXiv:1909.11258*, 2019.
- [3] M. Aksoy, S. Yanik, and M. F. Amasyali, "Reviewer assignment problem: A systematic review of the literature," *Journal of Artificial Intelligence Research*, vol. 76, pp. 761–827, 2023.
- [4] V. S. Pendyala, K. Kamdar, and K. Mulchandani, "Automated research review support using machine learning, large language models, and natural language processing," *Electronics*, vol. 14, no. 2, p. 256, 2025.
- [5] M. D. Khilji, "Features matching using natural language processing," *arXiv preprint arXiv:2303.12804*, 2023.
- [6] I. Ali, N. Mughal, Z. H. Khand, J. Ahmed, and G. Mujtaba, "Resume classification system using natural language processing and machine learning techniques," *Mehran University Research Journal of Engineering & Technology*, vol. 41, no. 1, pp. 65–79, 2022.
- [7] X. Li and T. Watanabe, "Automatic paper-to-reviewer assignment, based on the matching degree of the reviewers," *Procedia Computer Science*, vol. 22, pp. 633–642, 2013.
- [8] S. T. Boddu, S. Desu, and S. Puli, "Resume summarizer and job description matcher using natural language processing and spacy," *Interantional Journal Of Scientific Research In Engineering And Management*, vol. 7, pp. 1–11, 2023.
- [9] G. Narendra and S. Hashwanth, "Named entity recognition based resume parser and summarizer," *Int. J. Adv. Res. Sci. Commun. Technol*, vol. 2, pp. 728–735, 2022.